

Project Proposal & Proposed Solution Report

The proposal report aims to transform credit card approval decisioning using machine learning, boosting operational efficiency and decision accuracy. It tackles manual underwriting bottlenecks, promising faster processing, decreased loan defaults, and an enhanced customer application experience. Key features include a multi-algorithm machine learning model, rule-based delinquency overrides, and real-time interactive decisioning portals.

Project Overview	
Objective	The primary objective is to automate the credit card approval decisioning process by implementing advanced machine learning classification techniques, ensuring faster, unbiased, and more accurate risk assessments.
Scope	The project comprehensively covers the modeling pipeline (training 4 algorithms: Logistic Regression, Decision Tree, Random Forest, XGBoost), backend REST API endpoint development, compliance audit filters, and a stepped customer pre-qualification check wizard.
Problem Statement	
Description	Addressing operational bottlenecks and underwriting errors in manual credit screening workflows. Manual review is highly time-consuming, costly, and susceptible to cognitive bias and risk assessment mistakes.
Impact	Automating decisions resolves processing backlogs, decreases the bank's non-performing loan (NPL) ratios, and prevents credit score damage for customers, contributing to regulatory compliance and higher customer satisfaction.
Proposed Solution	
Approach	Employing automated machine learning classifiers (specifically Decision Tree selected on Highest Accuracy criteria) to preprocess and score applicant financial vectors in under 50ms.
Key Features	- Machine learning-based credit underwriting scoring (Decision Tree active model). - Real-time decision explainability (visual progress dial + positive/negative risk factor lists). - Session-persistent Analyst history logging table. - Compliance overrides automatically blocking delinquency status codes 1-5. - Soft-check pre-qualification check wizard bypassing credit bureau score drops.

Resource Requirements

Resource Type		
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i7 / AMD Ryzen 7, 8 Cores (local dev environment) or IBM Cloud Virtual Server
Memory	RAM specifications	16 GB DDR4 RAM
Storage	Disk space for data, models, and logs	512 GB SSD
Software		
Frameworks	Python frameworks	Flask (v3.1.3), Gunicorn (WSGI)
Libraries	Additional libraries	scikit-learn (v1.9.0), pandas (v3.0.3), numpy (v2.5.0), Chart.js (v4.4.1)
Development Environment	IDE	VS Code, PyCharm, Jupyter Notebooks
Data		
Data	Source, size, format	Synthetic credit approval dataset (2,500 records, CSV format), UCI Credit Approval Dataset (690 records, CSV format)